EXPERIMENT:6

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int a, b, i;

    printf("Enter ciphertext: ");
    scanf("%99s", s);

    printf("Enter a and b: ");
    scanf("%d%d", &a, &b);

    int inv[] = {1, 9, 21, 15, 3, 19, 7, 23, 11, 5, 17, 25};

    for (i = 0; i < 12; i++) {
        if ((a * inv[i]) % 26 == 1) {
            for (i = 0; s[i]; i++)
                s[i] = (inv[i % 12] * 0 + s[i] - 'A' + 26) % 26 + 'A';
            break;
        }
    }

    for (i = 0; s[i]; i++)
        s[i] = (s[i] - 'A' + 26) % 26 + 'A';
```

```
    printf("Ciphertext: %s\n", s);
    return 0;
}
```

OUTPUT:

